

Azaan Noman

azaannoman03@gmail.com

azaannoman.vercel.app

github.com/ieccclodd

EDUCATION

Seven Lakes High School | Katy, TX

Expected May 2028

Academic interests: Machine learning, autonomous systems, neuroscience, and biomedical engineering.

RESEARCH EXPERIENCE

sEMG Feature Transferability for Cross-User Hand-Gesture Classification

Co-author | BMES 2026 presentation forthcoming | [Research code](#)

- Evaluated **21 feature types** across **128 recording channels (2,688 predictors)** from **18 participants** to study hand-gesture recognition and generalization to unseen users.
- Compared discriminant analysis, support-vector machines, and random forests using within-user and **leave-one-subject-out (LOSO)** evaluation.
- Investigated feature transferability through a **378-fit** single-feature random-forest benchmark, ranking features by held-out balanced accuracy.

CALIBER: Context-Aware Calibration for Neural Network Adaptation

Research in progress

- Developing a hypernetwork-inspired calibration layer to adapt trained neural networks to new users with limited contextual data; planned evaluation covers accuracy, data requirements, and adaptation time.

Connectome-Driven Drone Autonomy

Research in progress

- Investigating how fruit-fly neural circuitry can inform autonomous drone navigation and control, connecting biological sensing and steering mechanisms with computational models.

TECHNICAL & PRODUCT EXPERIENCE

Worktree (tryworktree.com)

Co-founder | Product Strategy

- Compared AI developer tools including Conductor, Mosaic, and Hoplite, shaping Worktree's positioning around coordination between **human engineers and coding agents**.
- Translated concerns about fragmented context, overlapping agent work, and cross-device visibility into roadmap priorities for **shared workspaces, conflict awareness, and human oversight**.
- Helped shape a **local-execution strategy** using connected computers and lightweight message relays, weighing infrastructure cost against multi-device collaboration.

CalcQuest - Adaptive Calculus Learning Game

Junior Developer

- Developing a **machine-learning personalization algorithm** intended to adapt calculus practice to users' progress over time.
- Develop game design and visuals for an **AP Calculus AB/BC** practice game.

DroneLab - FPV Simulator & AI Experimentation Lab ([GitHub](#))

- Built a browser-based drone simulator using **React, Three.js, Rapier, and TensorFlow.js**, with practice missions, flight recording, and replay.
- Implemented behavior-cloning experiments and held-out evaluation against scripted and random controllers.

WORK EXPERIENCE

Yo Jing Swim Academy

Present

Swim Instructor | Competitive Swimming

- Teach children competitive swimming for **12 hours per week** in a paid role, using clear instruction and technique-focused feedback.

ENGINEERING & COMPETITIONS

RoboSpartans - FIRST Robotics Competition, Team 4639

2026 season

Electrical Engineering | Two-year team member

- Redesigned and implemented robot electrical layouts to accommodate major structural changes, reorganizing **power distribution, CAN wiring, motor-controller placement, sensor wiring, and cable management.**
- [2026 team results](#): **3rd of 31** in Dripping Springs qualifications; **District Event Finalist** and **Innovation in Control Award** recipient.

Destination Imagination - Engineering Challenge

2024

Global Finals | 3rd Place Globally (Team)

- Created the complete blueprint set and redesigned a classical trebuchet into a **transforming competition mechanism**, targeting greater mechanical effectiveness and efficiency.

Pumps & Pipes - ChemE Car

Grand Champion (Team)

- Designed a **chemical-reaction-controlled car from scratch**, integrating mechanical design, chemical formulations, and electrical controls.
- Troubleshoot wiring, motors, and sensors to resolve hardware integration issues before competition.

RECOIL-1: Bioinspired Elastic-Recovery Water Pump

- Designed a low-pressure pump in **AutoCAD** using a replaceable silicone membrane; investigating leaf-vein-inspired reinforcement to improve recovery and consistency of water delivery.

FPV & Racing Drones

- Built and flight-tested **three racing drones**, handling component selection, soldering, assembly, and advanced tuning.

LEADERSHIP & SERVICE

Hearts for Elders

Head of Chapter Management

- Coordinate volunteers and local teams across **6 continents**. Organization-wide impact: **1,150+ elders reached, 200+ volunteers, and \$10,000+ raised.**

Mecclabs

Mechanical Co-Director

- Design hands-on engineering camp curriculum across mechanical, electrical, biomedical, and chemical engineering; helped generate **nearly \$20,000** through educational programs.

Technology Student Association

Officer

- Help prepare **50+ competitors** for regional, state, and national events. Chapter results: **2nd among 125 schools** in the State School Award; **14 national qualifiers.**

SLHS Student Council

Officer

- Help lead a **600+ member** organization and support **\$100,000+ in annual fundraising** for meals and supplies donated to local shelters.

ADDITIONAL PROJECTS, SKILLS & ACTIVITIES

Skyes Travel: Built AI booking automation for a family travel business with \$50,000 in revenue.

Additional projects: Co-founder of [Morro](#) (adaptive workflow software) and [Polyshield](#) (AI-agent security).

Tools: MATLAB, React, Three.js, Rapier, TensorFlow.js, Vite, GitHub, AutoCAD.

Methods: Machine-learning evaluation, LOSO validation, behavior cloning, CAD, electrical layout, soldering.

Competitive climbing: **V7 bouldering**; speed-climbing personal best **7.4 seconds**; train approximately **4 hours per week.**